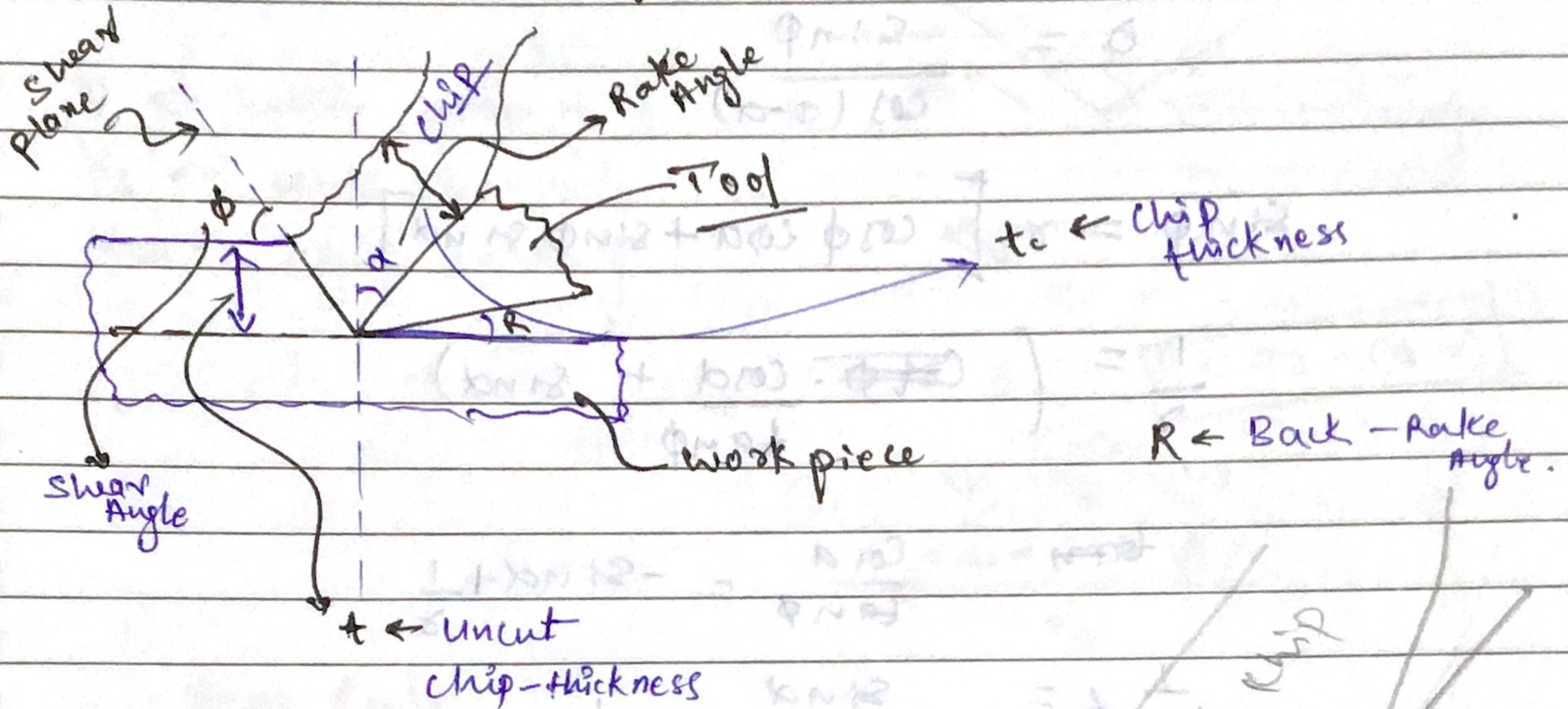
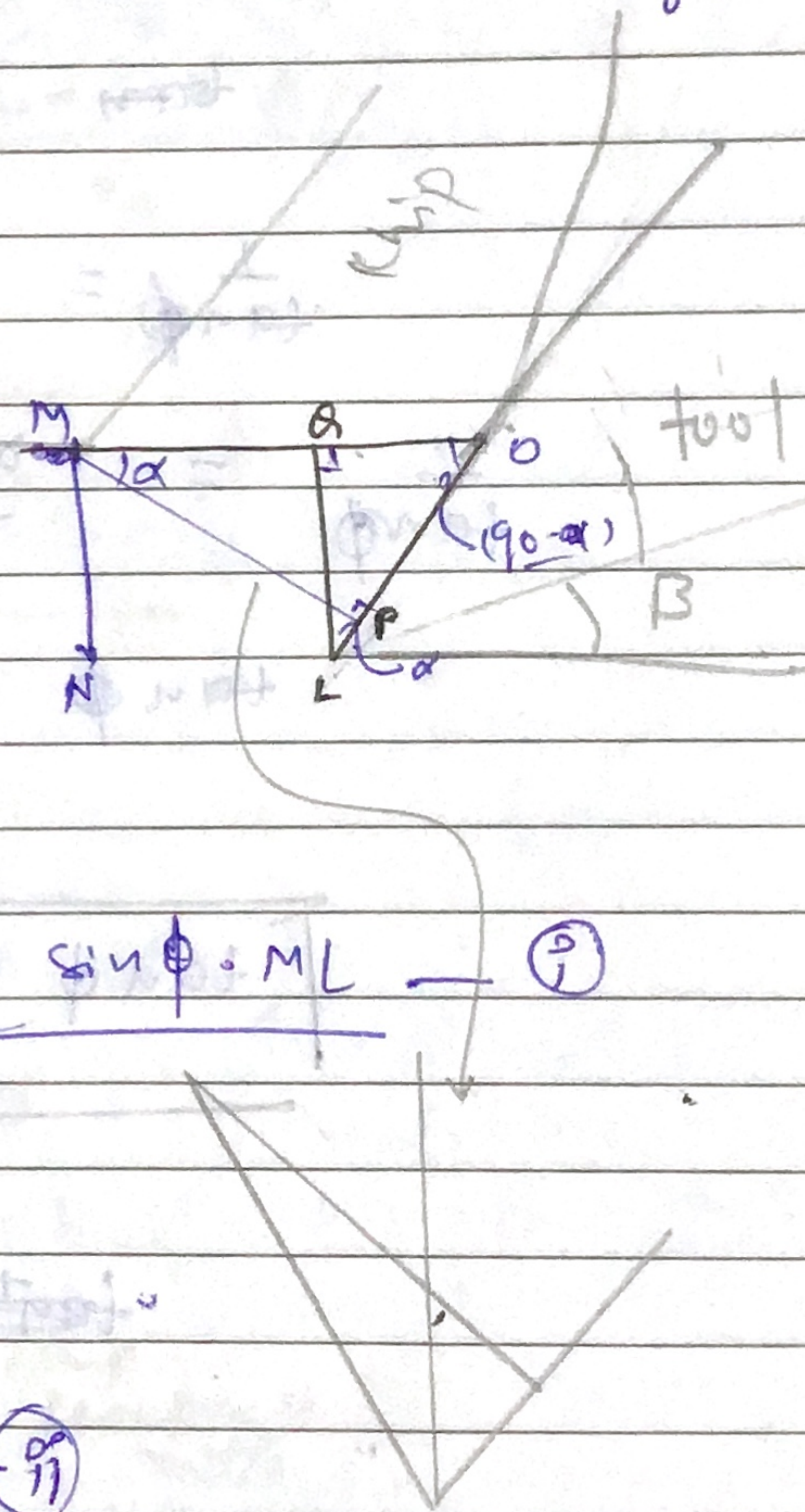


Orthogonal Cutting Mechanics



chip thickness ratio (γ)



In $\triangle MNL$

$$\sin \phi = \frac{QL}{ML}$$

$$\sin \phi = \frac{t}{ML} \Rightarrow t = \sin \phi \cdot ML \quad \text{--- (i)}$$

In $\triangle MLP$

$$\cos (\phi - \alpha) = \frac{MP}{ML} = t_c$$

$$t_c = ML \cos (\phi - \alpha) \quad \text{--- (ii)}$$

$$\gamma = \frac{t}{t_c} \Rightarrow \frac{\sin \phi}{\cos (\phi - \alpha)}$$

Shear angle

$$\gamma = \frac{\sin \phi}{\cos(\phi - \alpha)}$$

$$\sin \phi = \gamma [\cos \phi \cos \alpha + \sin \phi \sin \alpha]$$

$$\frac{1}{\gamma} = \left(\frac{\cos \alpha}{\tan \phi} + \sin \alpha \right)$$

$$\frac{\cos \alpha}{\tan \phi} = -\sin \alpha + \frac{1}{\gamma}$$

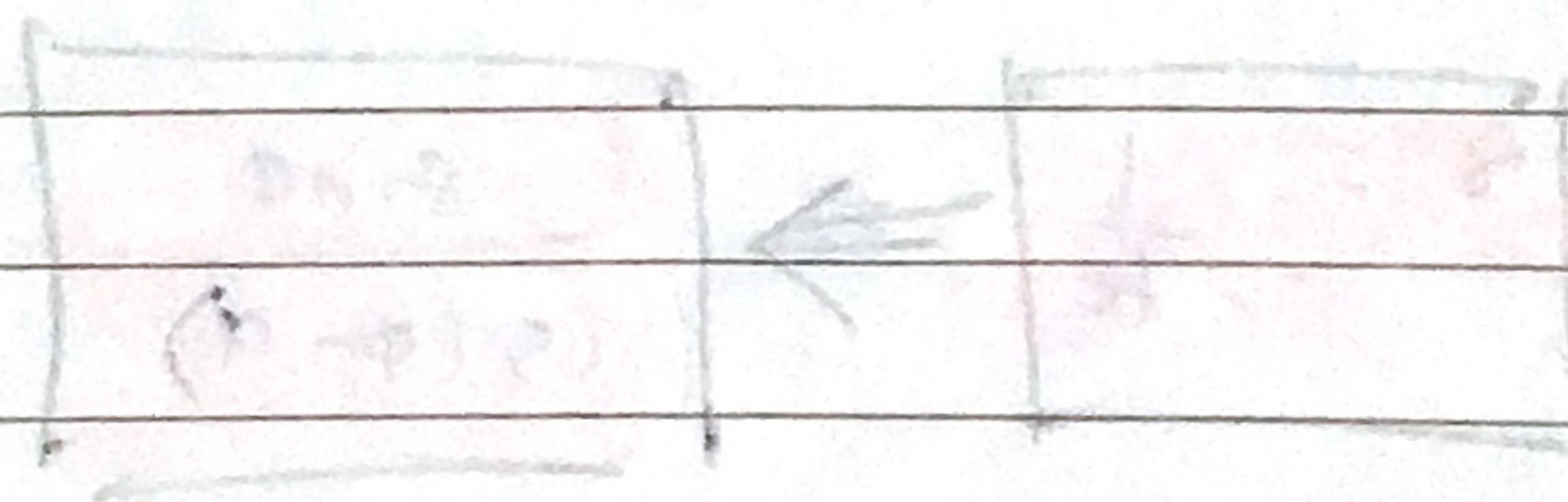
$$\frac{1}{\tan \phi} = \frac{\sin \alpha}{\cos \alpha} - \frac{1}{\gamma \cos \alpha}$$

$$\frac{1}{\tan \phi} = \frac{-\gamma \sin \alpha + 1}{\gamma \cos \alpha}$$

$$\tan \phi = \frac{\gamma \cos \alpha}{(-\gamma \sin \alpha + 1)}$$

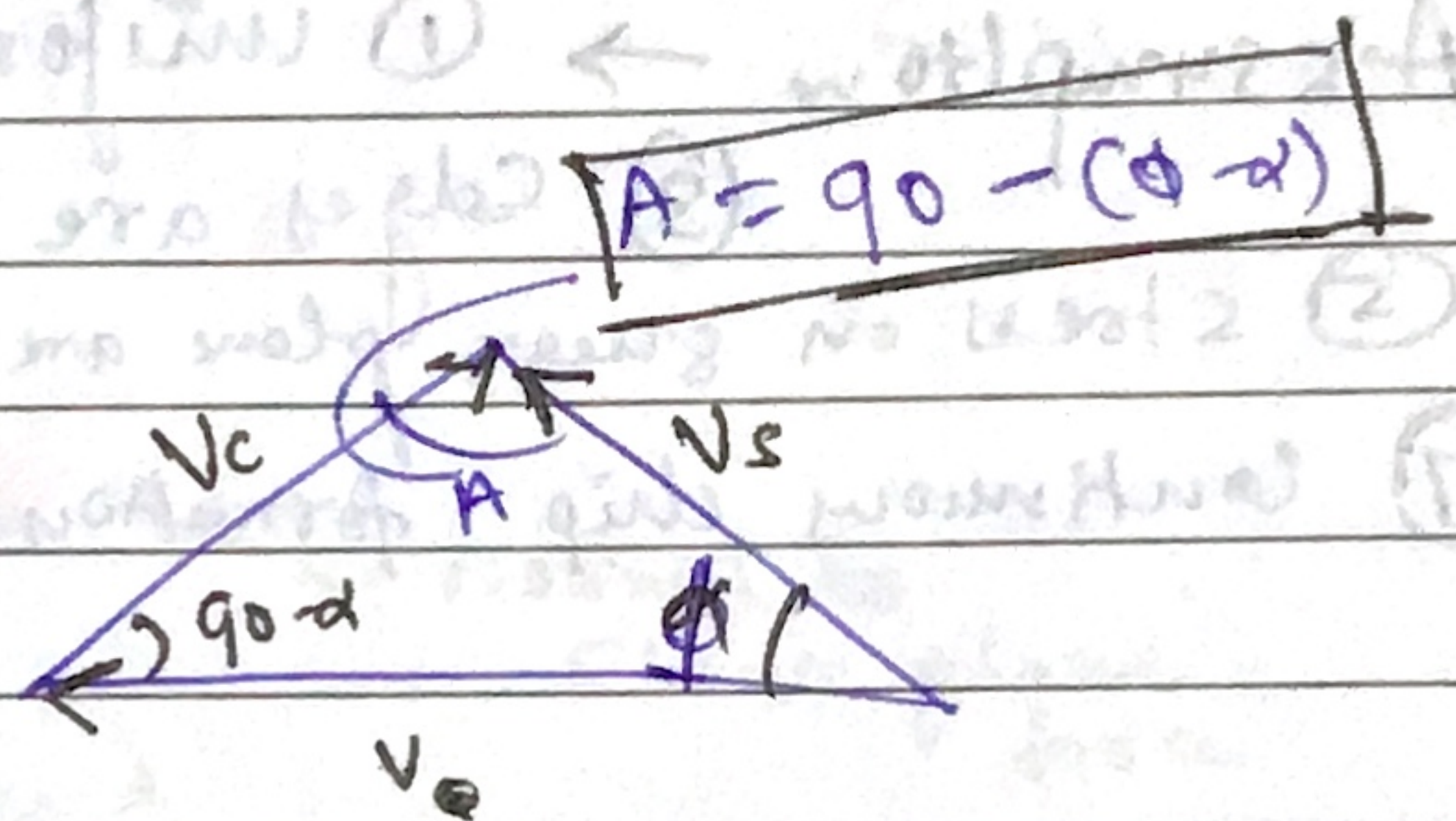
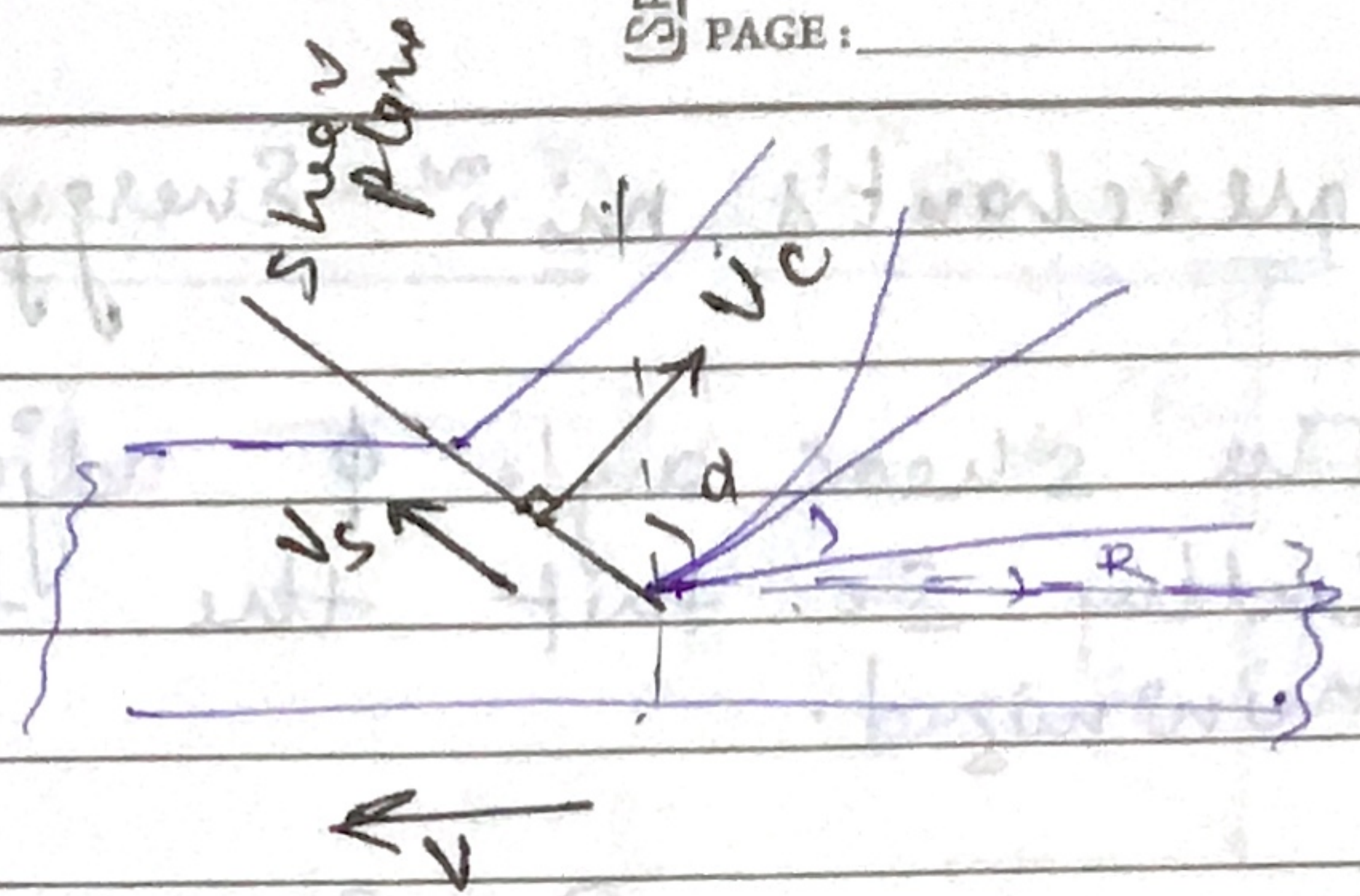
$$\tan \phi = \frac{\gamma \cos \alpha}{1 - \gamma \sin \alpha}$$

$$\phi = \tan^{-1} \left(\frac{\gamma \cos \alpha}{1 - \gamma \sin \alpha} \right)$$



Velocity

$V \leftarrow$ cutting velocity
 $V_s \leftarrow$ shear vel.
 $V_{ch} \leftarrow$ chip velocity



Sine Rule

$$\frac{V_c}{\sin \phi} = \frac{V_s}{\sin(90 - \alpha)} = \frac{V}{\sin(90 - (\phi - \alpha))}$$

If you note

$$\frac{V_c}{\sin \phi} = \frac{V}{\sin(90 - (\phi - \alpha))}$$

$$\frac{V_c}{V} = \frac{\sin \phi}{\cos(\phi - \alpha)}$$

$$\frac{V_c}{V} = r \quad \text{chip thickness ratio}$$

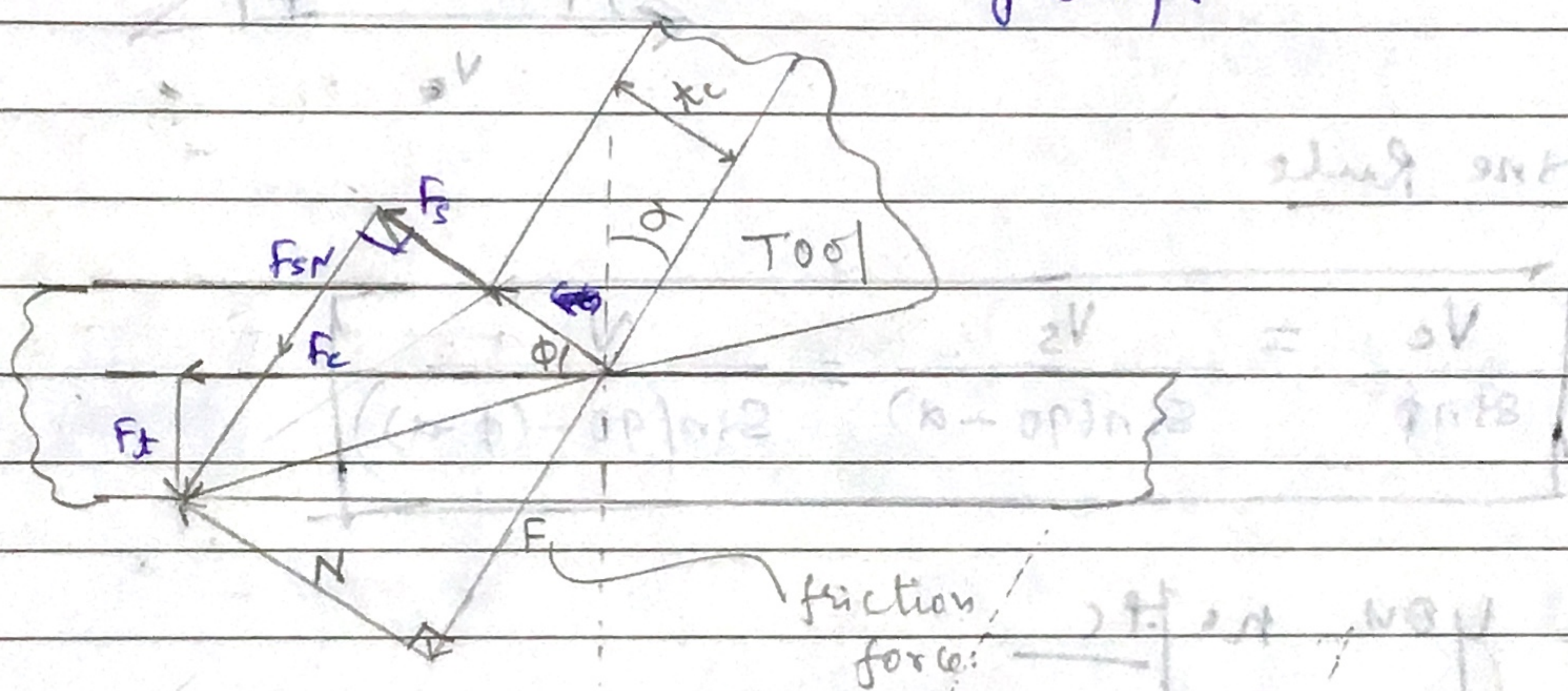
So,

$$r = \frac{V_c}{V} = \frac{t}{f_c}$$

Merchant's min^m Energy theory

The shear angle ϕ adjusts itself during its cutting so that the total cutting energy is minimized.

- Assumption $\rightarrow$
- ① uniform cutting vel.
 - ② tool is sharp
 - ③ edges are straight line
 - ④ tool width $\gg$ work width
 - ⑤ stress on shear plane are uniform.
 - ⑥ uncut chip thickness is const.
 - ⑦ continuous chip formation.
 - ⑧ No side flow of chips.



$F_t \leftarrow$ thrust force

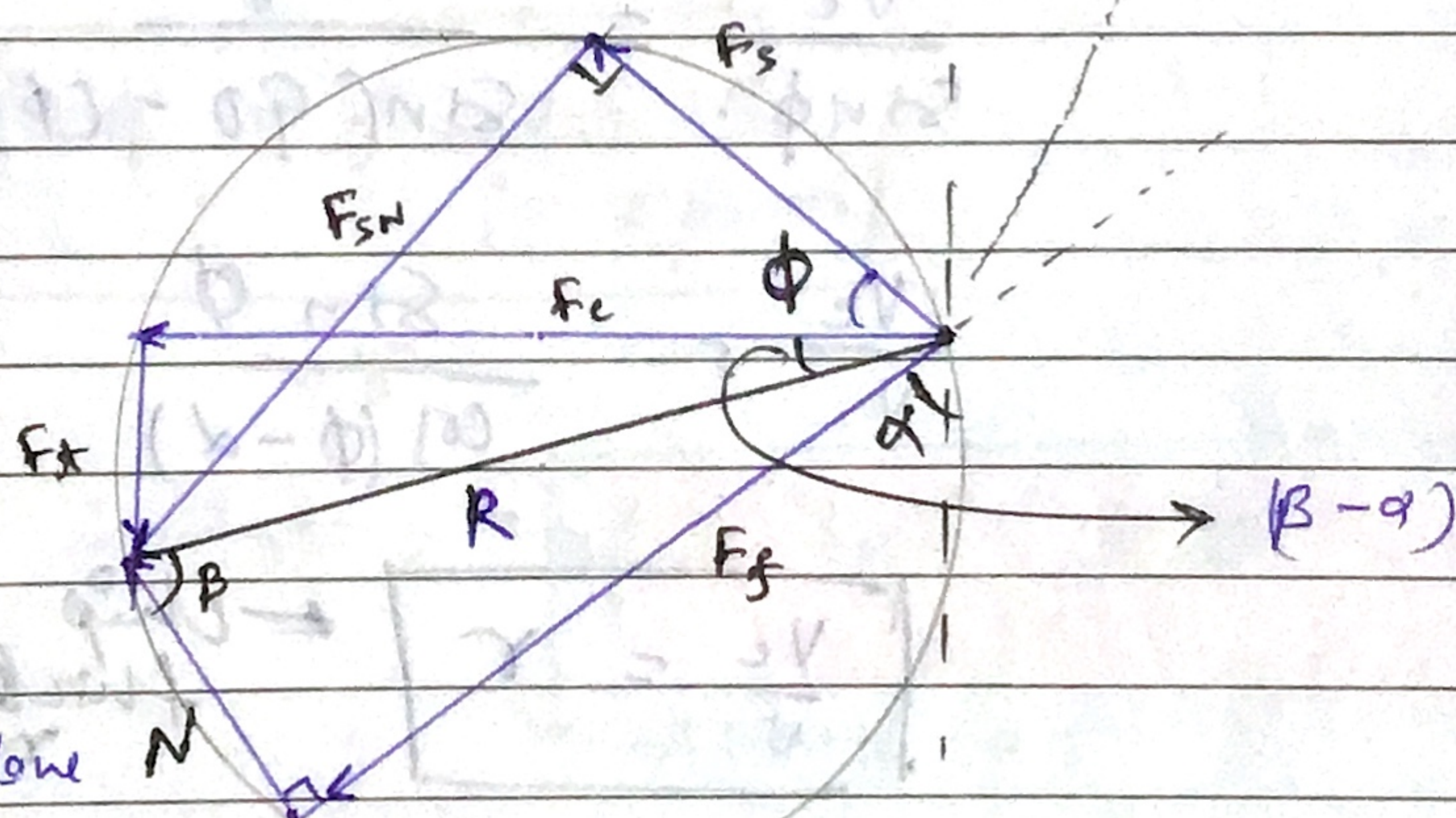
$F_c \leftarrow$ cutting force

$F_f \leftarrow$ friction force

$N \leftarrow$ Normal to

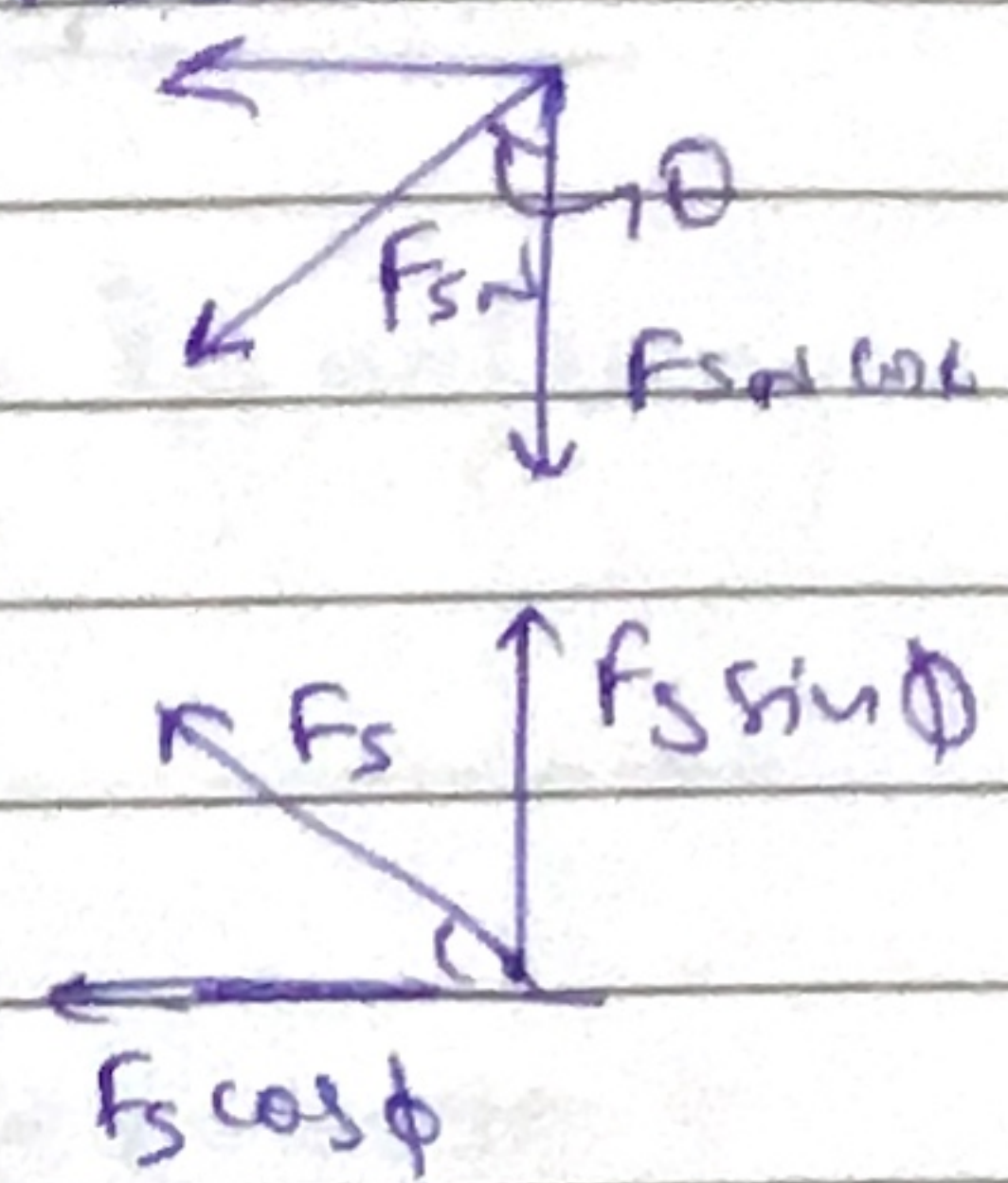
$F_s \leftarrow$ shear force

$F_{sN} \leftarrow$ Normal to shear plane



$$F_c = F_{SN} \sin \phi + F_s \cos \phi \longrightarrow \textcircled{i}$$

$$F_t = F_{SN} \cos \phi - F_s \sin \phi \longrightarrow \textcircled{ii}$$



$$\textcircled{i} \times \sin \phi + \textcircled{ii} \times \cos \phi$$

we get

$$F_{SN} = F_c \sin \phi + F_t \cos \phi$$

Normal to shear plane force

$$\textcircled{i} \times \cos \phi + \textcircled{ii} \times \sin \phi$$

we get

$$F_s = F_c \cos \phi - F_t \sin \phi$$

shear force

$$F_f = F_c \sin \alpha + F_t \cos \alpha$$

friction force

$$N = F_c \cos \alpha - F_t \sin \alpha$$

Normal

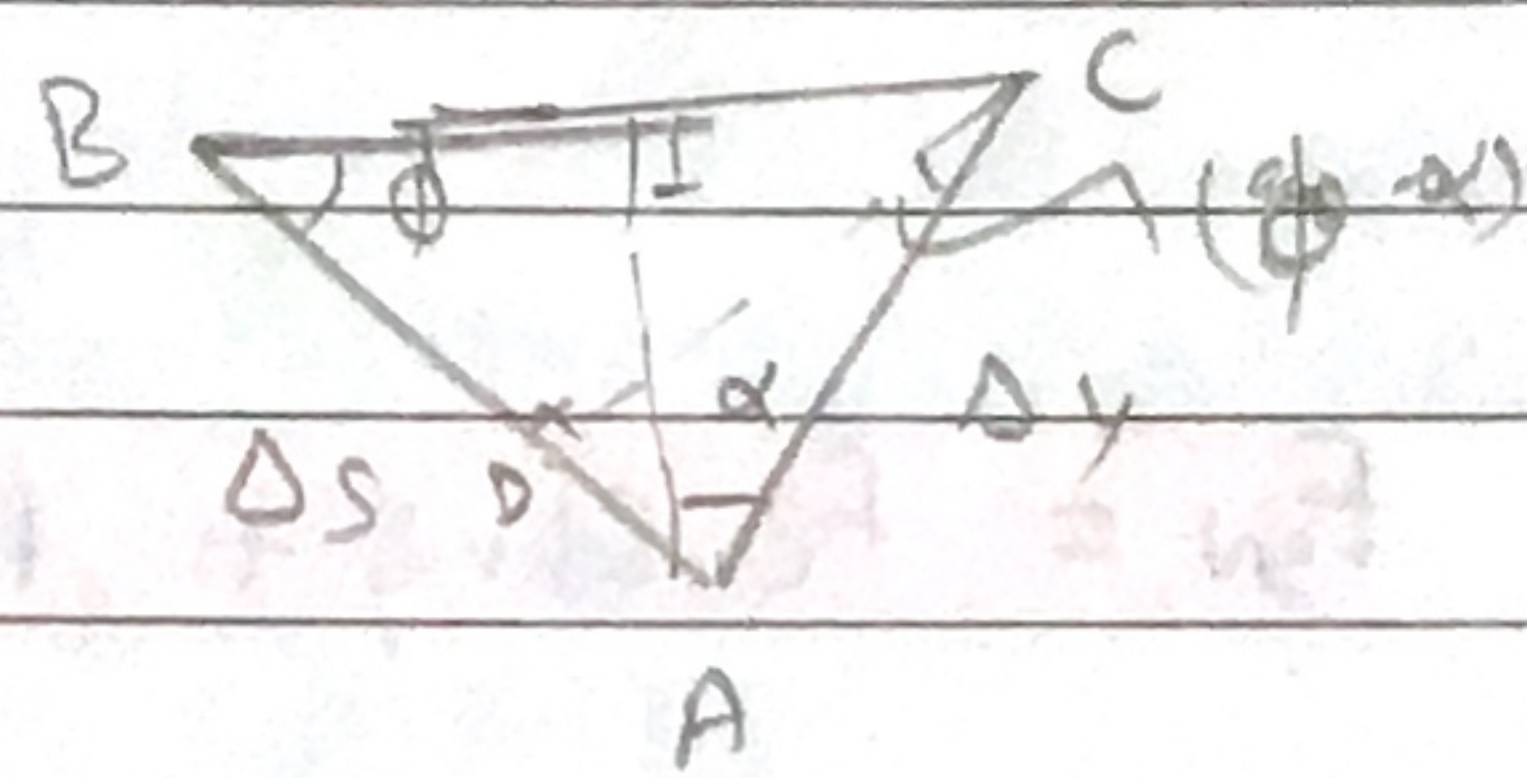
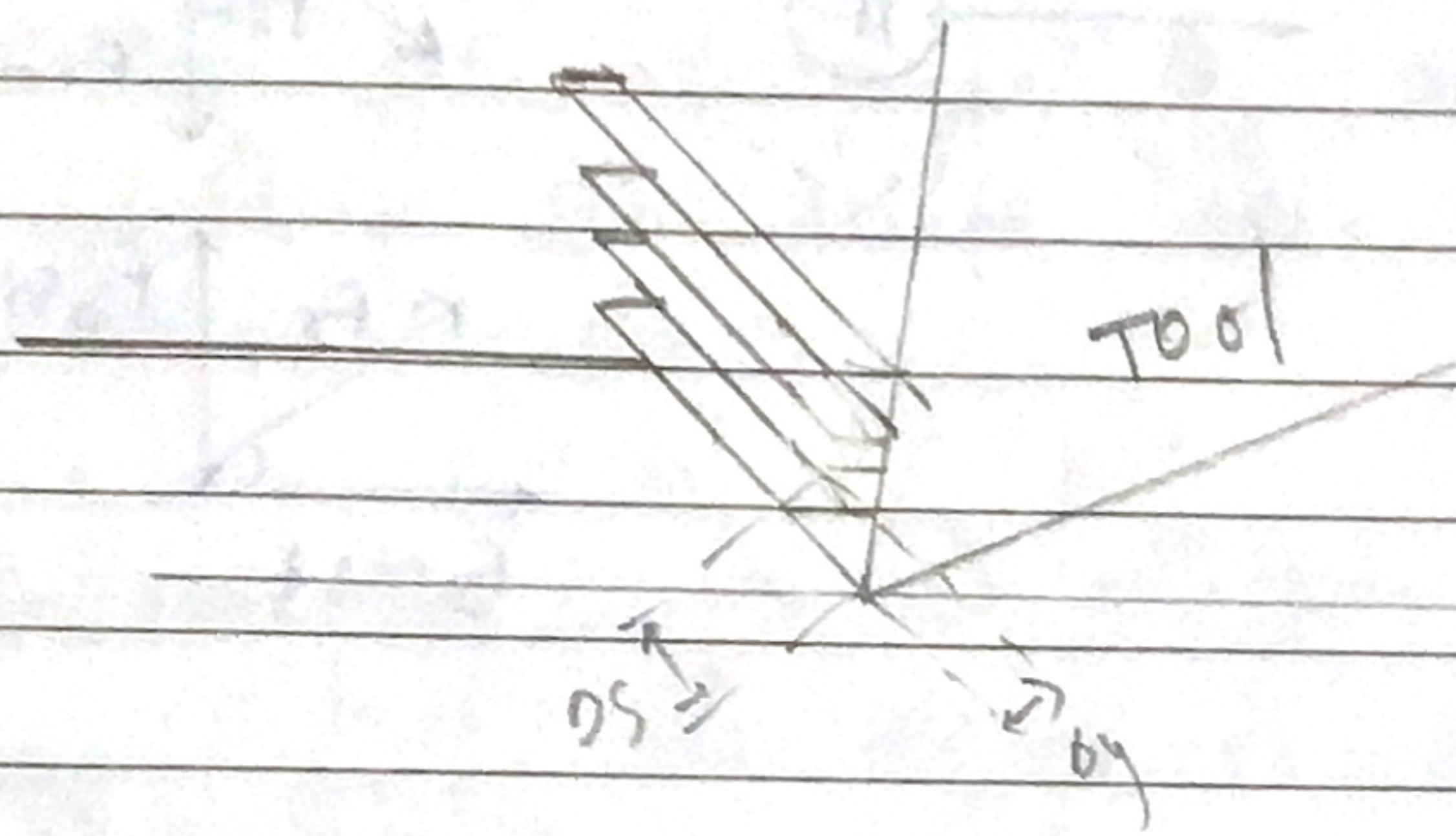
$$\mu = \tan \beta = \frac{F_f}{N}$$

frictional angle

$$\phi = 45^\circ + \frac{\alpha}{2} - \frac{\beta}{2}$$

shear angle

Stress & strain

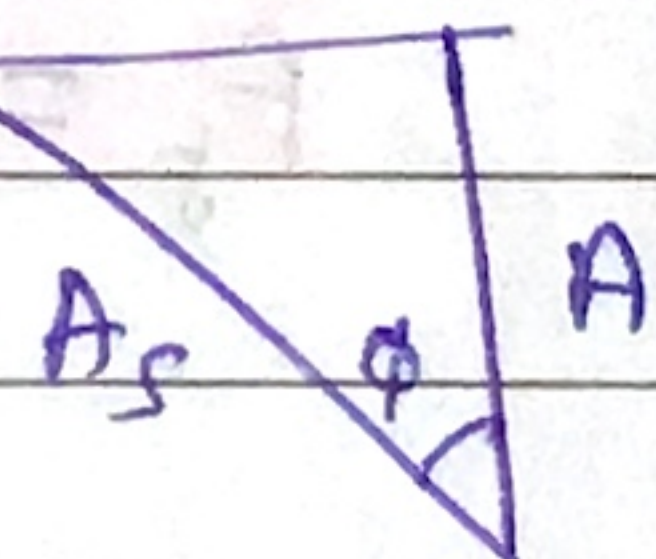
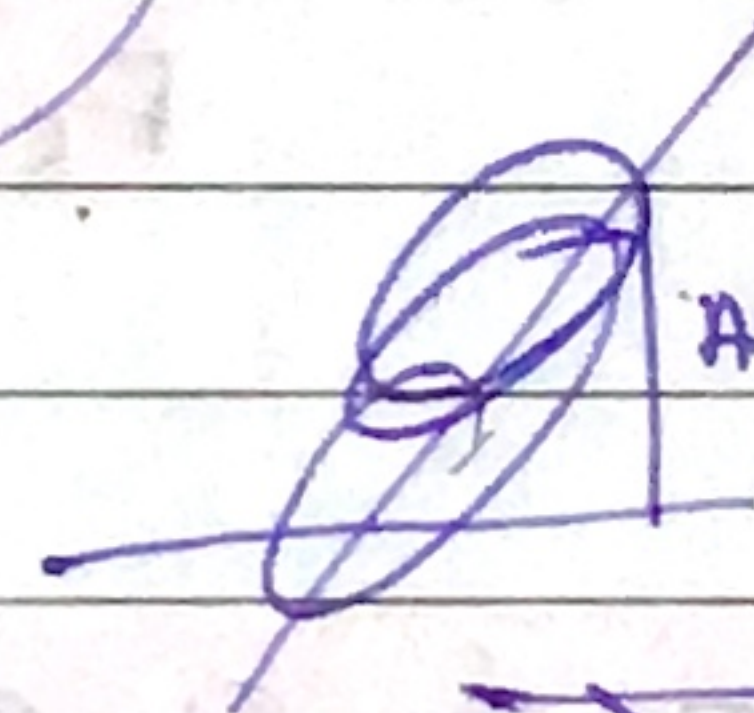


$$\tau = \frac{F_s}{A_s}$$

↑
Shear stress

$$A_s = \frac{A}{\sin \phi}$$

Shear plane Area.



$$\sigma = \frac{F_c}{A_s}$$

↑
Normal stress

↓
Strain

$$\epsilon = \frac{\Delta L}{L}$$

or

↓
Strain rate

$$\dot{\epsilon} = \frac{v}{\Delta y}$$

Dynamometry

Define
Used to measure cutting forces directly. Involving in the operations.

methods to measure

- ① deflection by a body
② strain induced in the body
③ pressure exerted on the body.

Dynamometer req.

- ① Sensitive enough to detect minor deflections.
- ② High rigid and stiff so it won't get itself disturbed.
- ③ Should get affected by each other while measuring diff. component (x, y, z) was multi channel dynamometer.
- ④ Stable under operations/ environment.
- ⑤ Should be easily calibrated.

① measuring the ~~strain~~ deflection → measures small displacement caused by force.

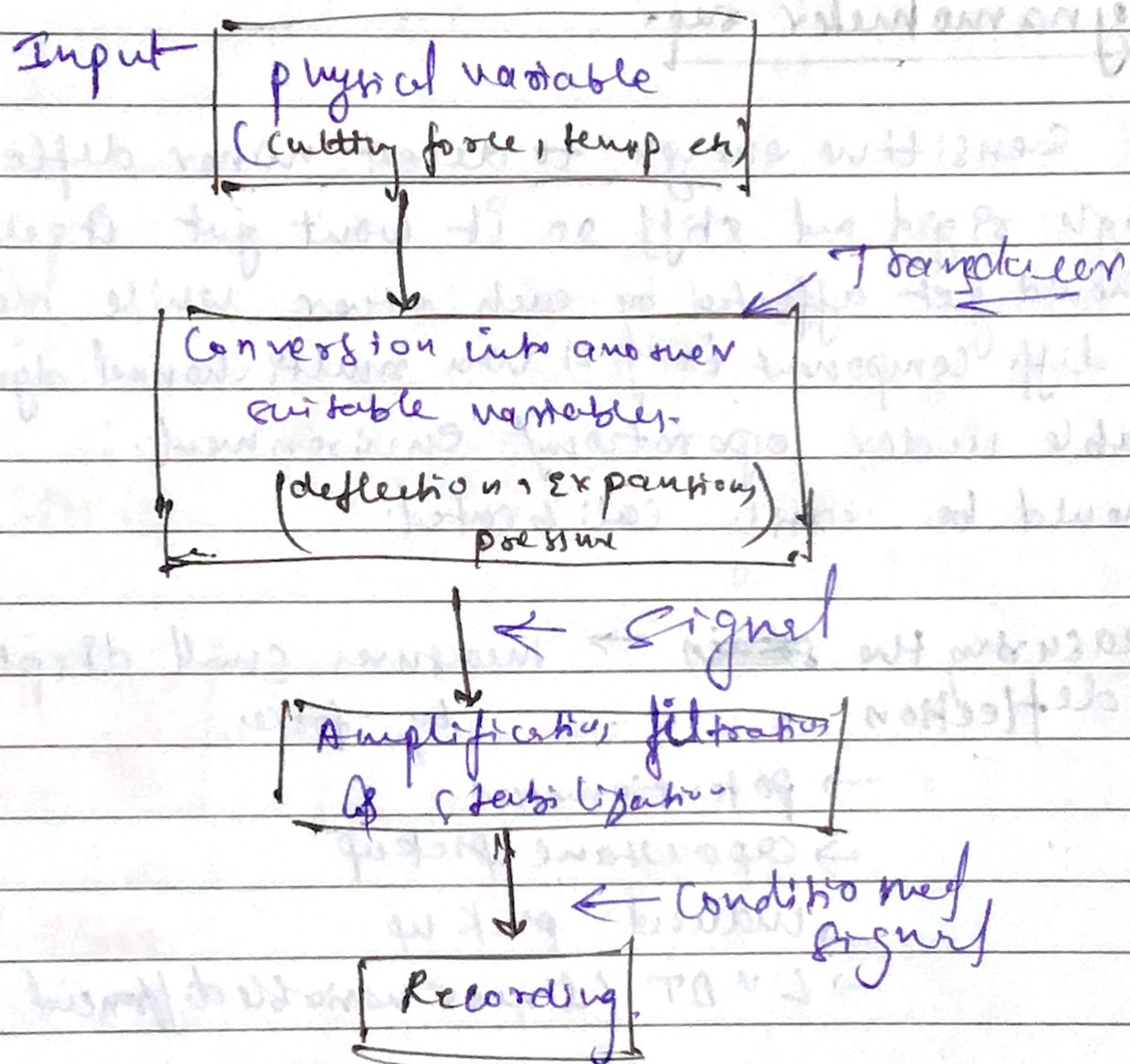
- potentiometer
- capacitance pickup
- induced pickup
- LVDT (linear variable differential)

② measuring strain → detects deformation due to applied force.

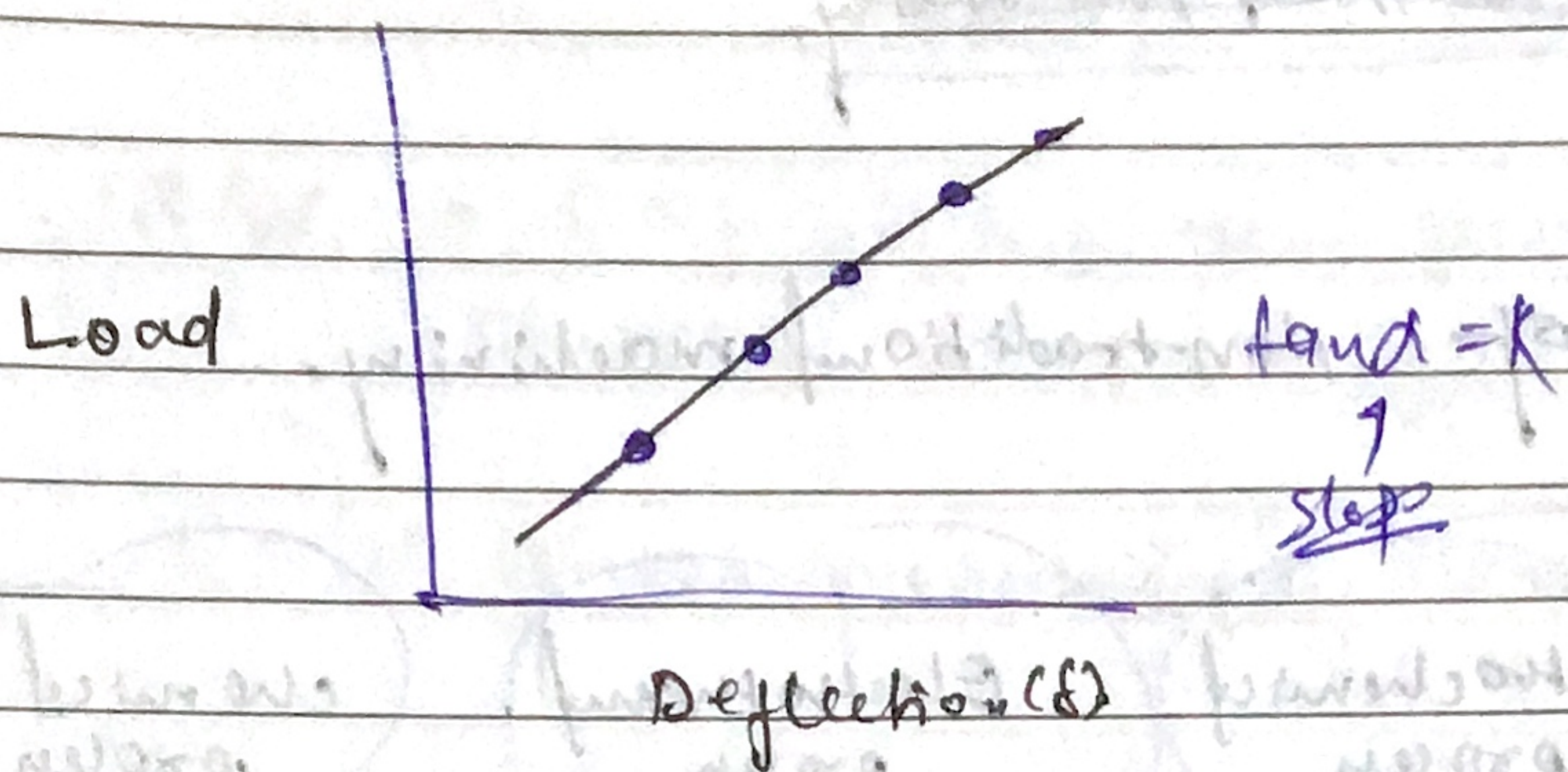
Strain gauges → full
half } → Bridges,
Quarter

③ measuring pressure → Suitable for high force applications
→ piezoelectric crystal → generates voltage in response to applied mechanical pressure.

procedure of dynamometry
(principle)



Transducers → convert mechanical / physical quantity into electric signals.



Non-Conventional machining.

Traditional machining → used direct contact and shear forces to machine
can only work for weaker or less hard materials.

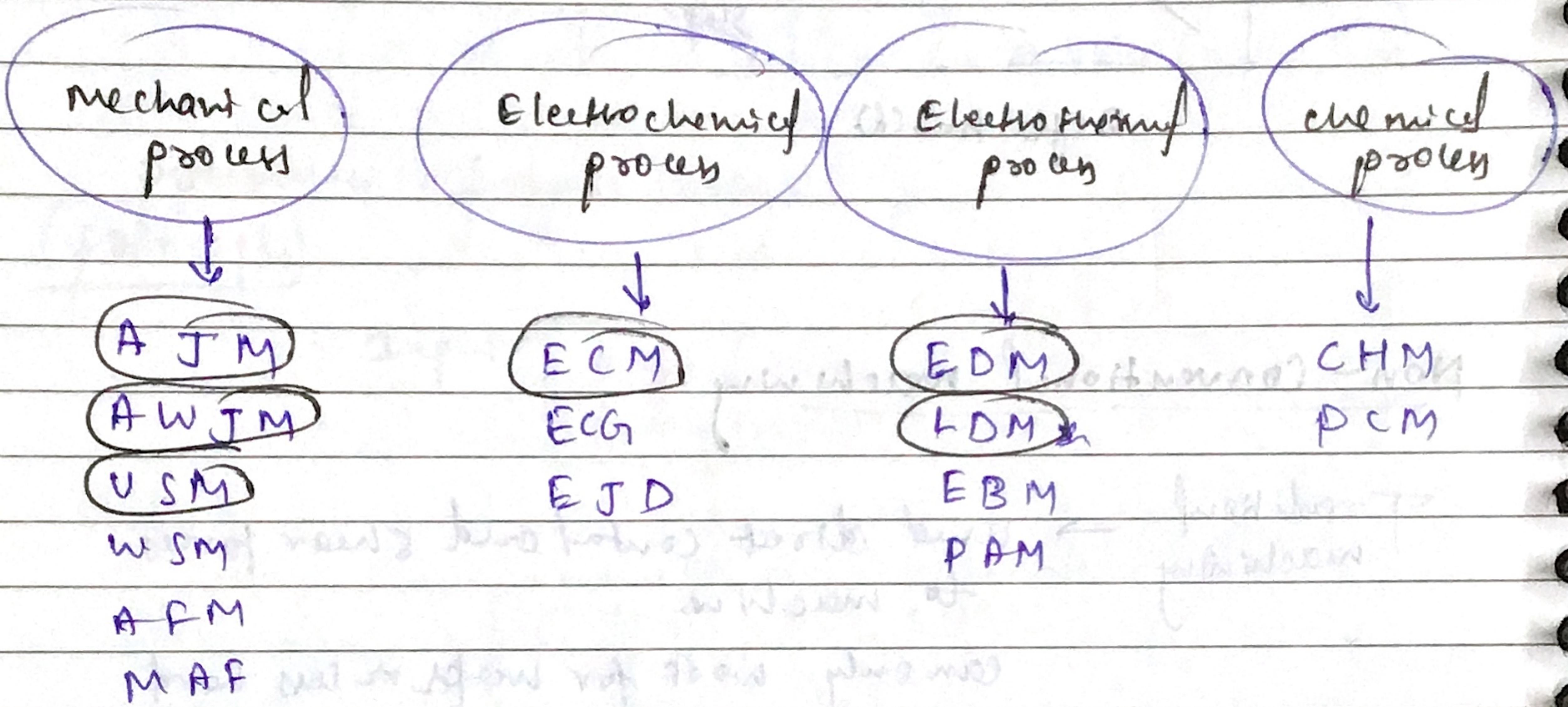
Non-traditional machining → Direct contact is not desired in most of operation.
Can work for → very hard materials
Complex shaped
High precision
difficult metals and composites.

Why we needed → to machine ultra hard material (carbides & ceramics)
for micro features,
to avoid tool wear (non-consumable)
to reduce thermal and mechanical stresses.

Advantages of non-traditional machining

Q

Classification of Non-traditional machining.



① Abrasive jet machining AJM

Abrasive → ~~hard~~ Rough particles that used to strike on sub through the surface of w/p to remove.

working → Abrasive particle mixed with gas strike with high speed to the surface.